DECK 5 OF 12 · FRC JAVA CRASH COURSE

Elastic: your robot on a dashboard

See live data, pick an auto and tune a value from the laptop

WPILib 2026 · Java · Command-based

OVERVIEW

What this deck covers

Read the robot

Send numbers from Java and watch them live in Elastic.

Pick an auto

Give the drive team an auto chooser on the laptop.

Tune a value

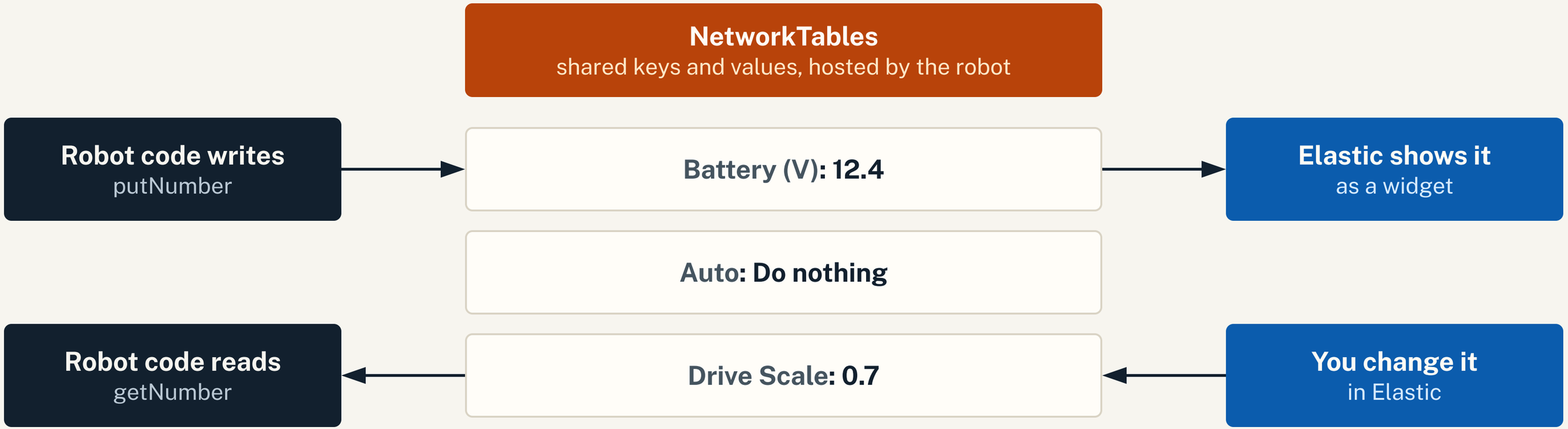
Change a number from the dashboard safely, and see what it does.

Keep your layout

Save the layout in your repo and load it from the robot.

WHAT A DASHBOARD IS FOR

NetworkTables: values shared by robot and laptop



Each line is a key and a value. Either side can write, and both see the change.

GETTING STARTED

Open Elastic and connect it to the robot

1. **Open Elastic**
   Driver Station gear tab: set **Dashboard Type** to Elastic. Or VS Code: WPILib logo, **Start Tool**, **Elastic**.

2. **Open the network settings**
   Click **Settings**, then the **Network** tab.

3. **At a robot with the Driver Station running**
   Set **IP Address Mode** to **Driver Station**. This is the default.

4. **Without the Driver Station, or in simulation**
   Team Number (10.TE.AM.2) with your **Team Number**, or **localhost (127.0.0.1)** for the simulator.

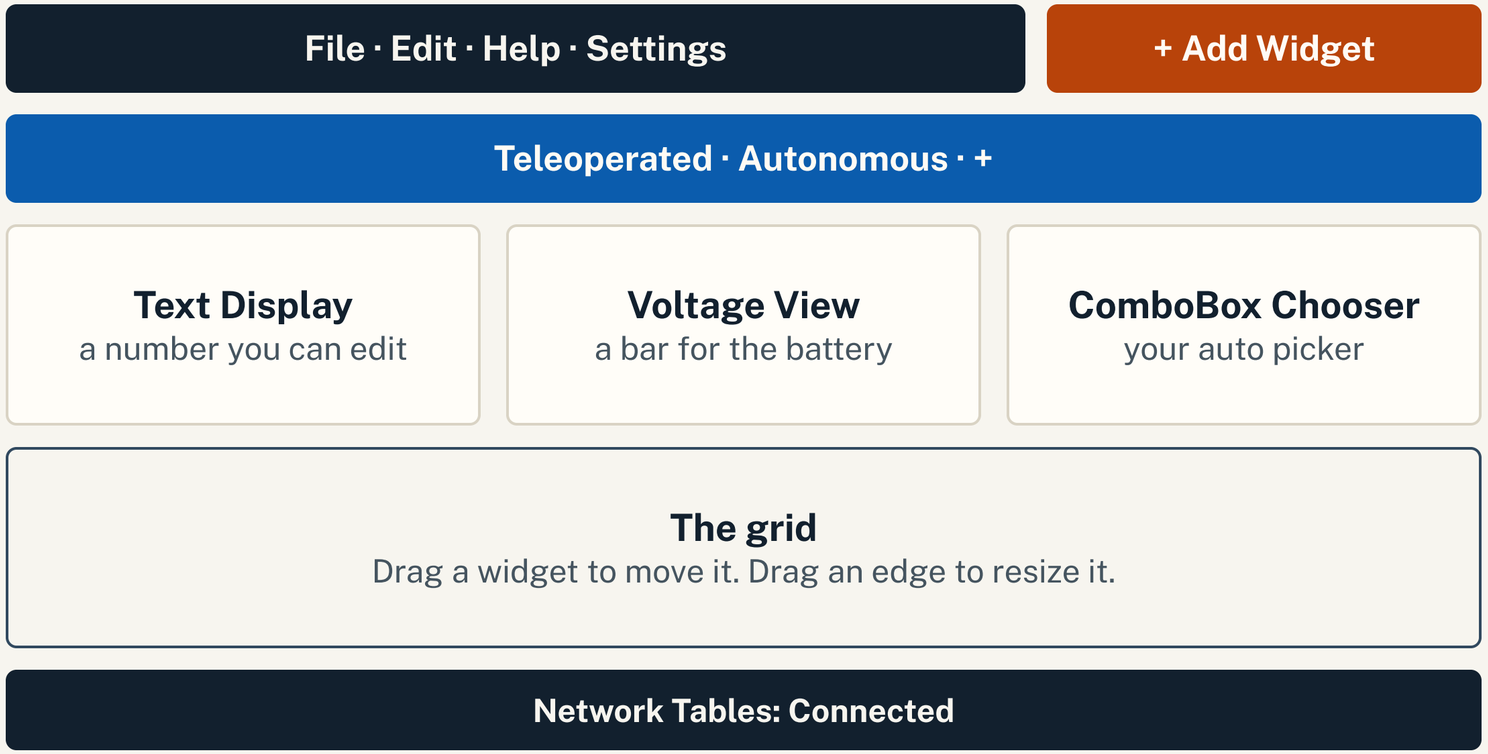
5. **Check the bottom of the window**
   Network Tables: Connected means Elastic has reached the robot.

Connecting to your robot

frc-elastic.gitbook.io

THE ELASTIC WINDOW

The window: tabs, a grid and the Add Widget button



Add Widget
Expand **SmartDashboard**, then drag a value onto the grid.

Right-click a widget
Show As changes its type. **Edit Properties** has the rest.

Right-click a tab
Rename, Duplicate or Close.

Edit, Lock Layout
Stops widgets moving at a match.

Schematic, not a screenshot.

SENDING DATA FROM JAVA

Send data with SmartDashboard

CANDriveSubsystem.java

```
  /** Once, from the constructor: items with a widget. */
  private void publishWidgets() {
1   SmartDashboard.putData(this); // shows the running command
2   SmartDashboard.putData("Drive Out 1 s", driveOut(1.0));
  }

  /** Every 20 ms, on its own. Send what someone will read. */
  @Override
3 public void periodic() {
    SmartDashboard.putNumber(
4       "Battery (V)", RobotController.getBatteryVoltage());
    SmartDashboard.putNumber(
        "Left Output (%)",
        leftLeader.getAppliedOutput() * 100.0);
  }
```

1. **A subsystem widget**
   It shows the running command, so name yours with withName.

2. **A command button**
   Click it to run the command on an enabled robot.

3. **Every 20 ms**
   Send a few useful values. Put the rest in logs.

4. **Units in the name**
   Battery (V) needs no explanation at a match.

2027: SmartDashboard is replaced by the Telemetry and Tunables APIs.

AUTONOMOUS

An auto chooser with SendableChooser

RobotContainer.java

```
  private final SendableChooser<Command> chooser =
      new SendableChooser<>();

  private void configureAutoChooser() {
1   chooser.setDefaultOption("Do nothing", Commands.none());
    chooser.addOption("Drive 2 s", drive.driveOut(2.0));
    chooser.addOption("Drive 4 s", drive.driveOut(4.0));
2   SmartDashboard.putData("Auto", chooser);
  }

  /** Robot.java calls this when auto starts. */
  public Command getAutonomousCommand() {
3   return chooser.getSelected();
  }
```

1. **A safe default**
   If nobody picks, the robot does nothing.

2. **Send it once**
   Drag Auto out of **SmartDashboard** in Add Widget.

3. **Read it late**
   Calling getSelected() here means the last pick wins.

2027: SendableChooser is replaced by Selectable.

COMMON MISTAKE

Live tuning: Enter disables an enabled robot

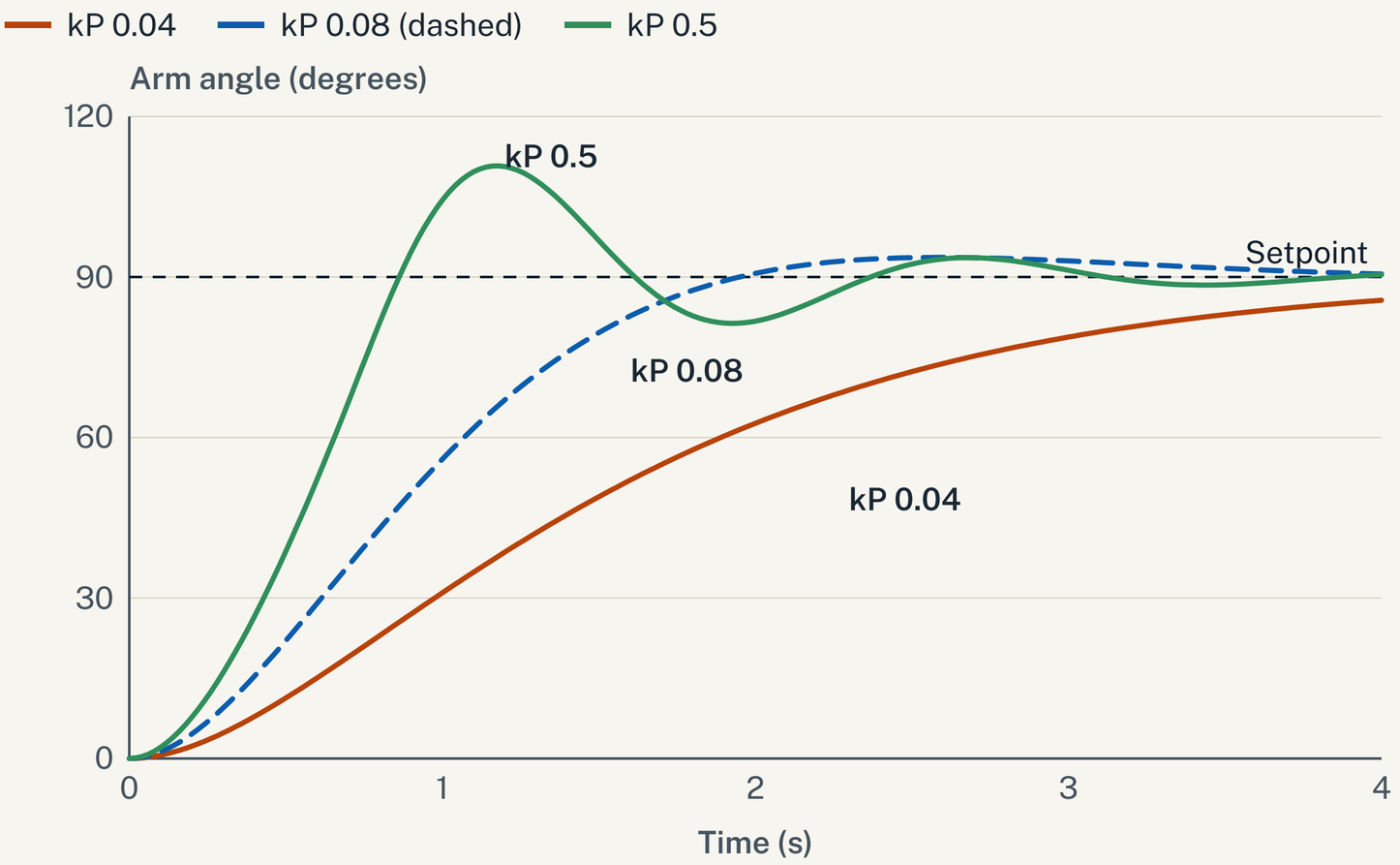
What you see	You type a new value into a box, press Enter, and the enabled robot stops.
Why it happens	Enter is the Driver Station shortcut for Disable, even while you are typing.
The fix	Robot enabled? Click the box’s submit button: **Show Submit Button** in **Edit Properties**.

```
SmartDashboard.setDefaultNumber("Drive Scale", DRIVE_SCALE); // once: only fills an empty box
DoubleSupplier driveScale =
    () -> SmartDashboard.getNumber("Drive Scale", DRIVE_SCALE); // every loop: reads the box
```


2027: The Tunables API replaces SmartDashboard for typed-in values.

CONTROL THEORY

Tuning a gain live from a graph



1. **One change at a time**
   Change one gain, then look. Change two and you cannot tell which one helped.

2. **A manual loop**
   Read the graph, decide, change the gain, then read it again.

3. **Too low, then too high**
   kP 0.04 reaches only 86 degrees in 4 s. kP 0.5 swings 21 degrees past.

4. **Send the gains**
   SmartDashboard.putData("Arm PID", pid)

BEST PRACTICE

A clean driver tab and a noisy debug tab

Driver tab: a quick glance

Battery and the auto chooser, drawn large.

Nothing you need to read twice.

Layout locked for matches.

Debug tab: everything

Motor outputs, gains and graphs.

Subsystem widgets and command names.

Keys with units, like Left Output (%).

BEST PRACTICE

Keep the layout on the robot

> **Put the layout file in your deploy folder, so it lives in the repo and on the robot.**

Save keeps it inside Elastic on one laptop. **Save As** writes a .json file you can commit. **Download From Robot** gives any laptop the same layout.

```
// In Elastic: File, Save As, into src/main/deploy/elastic-layout.json. Then deploy.
// Elastic's Download From Robot looks on port 5800, so start this once at startup.
private void serveLayout() {
  WebServer.start(5800, Filesystem.getDeployDirectory().getPath());
}
```

ELASTIC EXTRAS

Driver notifications and tab switching

RobotContainer.java

```
1 private final Elastic.Notification warning =
      new Elastic.Notification(
2         Elastic.NotificationLevel.WARNING,
          "Low battery",
          "Swap it.");

  private void configureDriverAlerts() {
    new Trigger(this::batteryLow)
3       .debounce(LOW_BATTERY_SECONDS)
        .onTrue(once(() -> Elastic.sendNotification(warning)));
    RobotModeTriggers.autonomous()
4       .onTrue(once(() -> Elastic.selectTab("Autonomous")));
    RobotModeTriggers.teleop()
4       .onTrue(once(() -> Elastic.selectTab("Teleoperated")));
  }
```

1. **Copy the helper**
   Add Elastic.java from Elastic's repo to your util folder.

2. **A popup**
   A level, a title and text. It shows for 3 s by default.

3. **Wait a moment**
   The battery must stay low for 2 s, so a quick dip is ignored.

4. **Switch tabs**
   selectTab takes a tab name. No match means no change.

COMMON MISTAKE

Six symptoms and where they come from

What you see	Why	Fix
A typed value snaps back at once	putNumber in periodic writes over the box every 20 ms	setDefaultNumber once, then getNumber
A typed value is gone after a code restart	NetworkTables starts empty and Elastic does not re-send it	Copy good values into Constants.java
A widget is blank	An old layout: the key was renamed, or never sent	Remove it and drag in the new key
A number never changes	The code sent it once, from a constructor	Send it from periodic
Every widget looks dimmed	Elastic is disconnected	Read the bottom bar, then IP Address Mode
The chooser shows a red mark	The robot lacks your pick. In 2026.1.1, Show As on a chooser stops it sending	Leave it as ComboBox Chooser

EXERCISE

Build a driver tab and save it

1. Add the Battery (V), Left Output (%) and Auto code, then deploy or simulate.
2. Click **Add Widget**, expand **SmartDashboard**, and drag Battery (V) and Auto to **Teleoperated**.
3. Right-click Battery (V), then **Show As**, **Voltage View**.
4. Click **+**, rename the new tab Debug, and drag Left Output (%) onto it.
5. **File**, **Save As**, into src/main/deploy/elastic-layout.json.

Done when: Auto shows a green check, and the file is in your deploy folder.

NEXT SEASON

What changes for 2027

In 2026	Changing for 2027
Older tutorials show Shuffleboard	Removed for 2027, so those steps do not apply
Elastic comes with WPILib	One of two recommended dashboards, with AdvantageScope
SmartDashboard.putNumber and putData	Telemetry and Tunables APIs
SendableChooser	Selectable

Everything in this deck is correct for 2026. WPILib 2027 is still in alpha, so details may change.

RECAP

What this deck covered

- Sending data with SmartDashboard and viewing it in Elastic.
- An auto chooser with a safe default.
- Tuning with the submit button, never Enter on an enabled robot.
- Changing one gain at a time and watching the graph.
- Keeping the layout in the deploy folder, so it lives on the robot.

Next: Deck 6 tunes an arm with PID and feedforward.

REFERENCE

Official docs



Elastic docs

frc-elastic.gitbook.io



Layouts from the robot

Remote Layout Downloading



Notifications and tabs

ElasticLib



NetworkTables

docs.wpilib.org